# RITHWIK RACHARLA

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## OBJECTIVE

Seeking a Software Engineering internship to apply strong foundations in data structures, algorithms, backend development, and full-stack engineering to build reliable software systems.

## EDUCATION

**B.Tech in Computer Science and Engineering** – VNR VJIET Expected 2028
CGPA: **8.93/10**

## SKILLS

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|---|---|
| **Programming Languages** | Java, Python, C++, JavaScript |
| **Core Concepts** | Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks |
| **Development** | React, Node.js, Express.js, REST APIs, JWT, MongoDB, MySQL, SQL |
| **Tools & Deployment** | Git, GitHub, VS Code, Vercel, Render, Sentry, PostHog |

## PROJECTS

**MedSafe – Medication Safety Checker** GitHub *Python, FastAPI, Groq, Pytest*

- Built medication-safety research prototype: parses prescription text, normalizes drug names, and flags drug–drug interactions, dose issues, and allergy conflicts via rule engine.
- Designed conservative pipeline where evidence-backed rule results stay authoritative; optional Groq explanations are non-authoritative, so LLM output cannot override validated checks.
- Sourced interaction and labeling data from DDInter and DailyMed with alias normalization; Pytest suite covers parsing, rule validation, and explanation behavior ([N] tests).

**Distill – AI News Digest** GitHub – Live *Node.js, transformers.js, Groq, Telegram/WhatsApp APIs, Vercel*

- Built personalized daily AI-news digest delivered on Telegram (LIVE) and WhatsApp (in testing): RSS fetch, interest-based filtering, Groq LLM summaries with "why read this" notes.
- Designed dedup pipeline around embedding-similarity clustering (MiniLM) rather than exact-match comparison, handling outlets phrasing the same event differently; confidence-tier logic separates single-source claims from multi-source-verified stories.
- Added multi-user support via Clerk Google OAuth and one-time-code Telegram linking. Deployed on Vercel with Sentry and PostHog monitoring.

**Library Management System** GitHub – Live *JavaScript, Node.js, Express.js, MongoDB, REST API*

- Built role-based (Admin/Faculty/Student) library system managing a 956-book catalog with live search and borrow/return logs.
- Secured endpoints with JWT middleware. Automated overdue fines and Web Push due-date alerts.
- Designed per-copy (not per-title) availability tracking with admin audit trail and student–faculty knowledge request module.

## COMPETITIVE PROGRAMMING

- Solved 500+ DSA problems across LeetCode (200+), Codeforces, CodeChef, SmartInterviews (200+).
- Contest maximum ratings: LeetCode **1672**, CodeChef **1414 (2⋆)**.

## CERTIFICATIONS & TRAINING

- HIVE – SmartInterviews, Data Structures & Algorithms.
- CSI BuildWithRAG Workshop – Built a RAG application on sports data.
- **Industrial Training** – MERN Stack (12 months), VNR VJIET; Agentic AI (2 months), VNR VJIET.